

Web Services Client Application

CS104D — Course Project Requirements Specification (with UI Architecture)

1. Project Overview & Summary

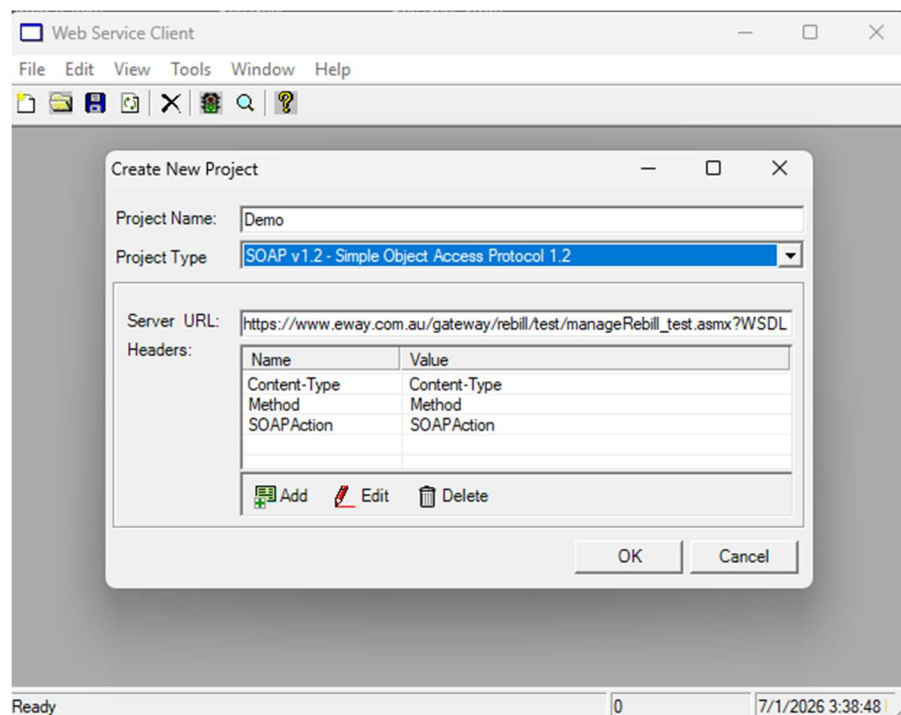
The objective of this project is to develop a robust Web Service Client application using Microsoft Visual Studio .NET. The application acts as an integration layer enabling business-to-business (B2B), application-to-application (A2A), and centralized data provider systems. It provides the core capability to dynamically create, locate, configure, and invoke remote Web Service methods using standard SOAP request-response structures over HTTP/HTTPS protocols.

2. Architectural Framework & User Experience

The application uses a Multiple Document Interface (MDI) architecture, in which the parent window manages multiple active project worksheets. Each separate workspace/project represents an independent Web Service definition instance.

Figure 1: Application Architecture & MDI Interface Layout Mockup

Below is the reconstructed layout diagram corresponding to the original embedded paintbrush interfaces, outlining the structured window configuration:



3. Step-by-Step Implementation Milestones

Milestone 1: Application Frame & Navigation Shell (Assignment 1)

Construct the primary MDI Parent window layout, including foundational menu components, global toolbar layouts, and structural feedback metrics:

- **File Menu:** New, Open, Close, Save, Save As, Save All, Project Definition, Preferences, Recent Project, Exit.
- **Edit Menu:** Copy, Paste.
- **View Menu:** Refresh.
- **Tools Menu:** Submit, Integration Broker, Search UDDI Registry.
- **Windows Menu:** Cascade, Tile Vertically, Tile Horizontally.
- **Help Menu:** Help, About.

Milestone 2: Configuration Modeling & State Persistence (Assignment 2)

Establish unified local state handling to maintain application rules and operational user profiles safely across active application runs:

- **Settings Model:** Implement an explicitly managed object blueprint enclosing global flags for General Settings, HTTP Proxy configs, and operational Security rules.
- **XML Serializer Core:** Leverage the .NET XmlSerializer workspace to seamlessly parse, persist, and deserialize active runtime values to and from a local disk file called settings.xml.

Milestone 3: Data Object Modeling & Document Framework (Assignment 3)

Design strongly typed core model frameworks capable of expressing deep schema settings, mapping them directly to underlying disk files:

- **Core Schema Structures:** Generate explicit internal object frameworks for WS_PROJECT_DEFN, WS_PROJECT_TYPE, WS_PROJECT_COOKIE, WS_PROJECT_HDR, WS_MESSAGE_TYPE, and WS_SETTINGS.
- **Stream Processing:** Build operational reading/writing hooks utilizing XmlDocument models, Stream wrappers, or standard XML serializers to preserve project workspaces dynamically.

Milestone 4: Custom UI Controls & Network Transport Engine (Assignment 4)

Establish dual raw/tree rendering panels for complex message construction while implementing the foundational outbound connection system:

- **Dual Rendering Strategy:** Map target XML datasets simultaneously to a RichTextBox container for explicit script adjustments and a structured TreeView layout for hierarchy-focused operations.

- **Transport Core:** Establish secure client-to-server interaction pipelines utilizing standard .NET HttpRequest and HttpResponse components to manage the SOAP exchange workflow safely.

Milestone 5: Project Definition Dialog Subsystem (Assignment 5)

Build a modular configuration window to safely record key network targets and contextual parameters:

- **Captured Parameters:** Project Name, Project Type (SOAP / WSDL / HTTP Request), Server Target URL, Custom Request Headers, SOAPAction declarations, and transport HTTP Headers.

4. Technical Feature Map Matrix

Functional Subsystem	Core .NET / Architectural Components	Status / Milestone
MDI Workspace Shell	MDI Form, MenuBar, ToolBar, StatusBar Controls	Completed (Milestone 1)
Configuration Framework	System.Xml.Serialization, Settings Class, settings.xml	Completed (Milestone 2)
Project Data Modeling	WS_PROJECT_DEFN, WS_PROJECT_TYPE, XmlDocument, File I/O Streams	Completed (Milestone 3)
Interactive Data Panels	System.Windows.Forms.RichTextBox, TreeView Controls	Completed (Milestone 4)
Networking Infrastructure	System.Net.HttpWebRequest, HttpResponse, SOAP Validation	Completed (Milestone 4)
Project Configuration	Modal Form Dialog (Project Type, URL, SOAPAction, Headers)	Completed (Milestone 5)